

6.2 Net Present Value, Internal Rate of Return and Real Options

Guiding Questions

1. Why is a dollar today worth more than a dollar later?
2. How do you convert future cash flows into present value?
3. What does a positive NPV tell you?
4. What does IRR represent?
5. Why might NPV and IRR sometimes give different signals?
6. What does real options analysis add that standard appraisal methods miss?

What will you learn?

- What NPV measures and why it matters.
- How to calculate present value of future cash flows.
- What IRR means and how it is interpreted.
- How to use Excel to calculate NPV and IRR.
- Why NPV is usually preferred when results conflict.
- Why real options analysis is needed beyond standard NPV and IRR.

NPV and IRR are discounted cash flow tools used to decide whether a project creates value. They recognize that money received earlier is worth more than money received later because money today can be invested.

Net present value

NPV compares the present value of all future cash inflows with the initial investment. If NPV is positive, the project is expected to add value. If NPV is negative, the project is expected to destroy

value.

$$NPV = \sum_{t=0}^n \frac{CF_t}{(1+r)^t}$$

Where: CF_t = cash flow in period t ; r = discount rate; and t = time period.

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Net Present Value and Internal Rate of Return



Internal rate of return

IRR is the discount rate that makes NPV equal to zero. It is often interpreted as the implied annual return generated by the project. If the IRR is above the required return, the project is usually acceptable.

NPV and IRR in project appraisal

NPV and IRR are both used to evaluate whether a project is financially attractive. They are related because both depend on discounted cash flows, but they are not identical. NPV measures value

added in dollar terms, while IRR measures the implied percentage return generated by the project.

The core logic is simple:

NPV asks: "How much value does this project create today?"

IRR asks: "What rate of return does this project generate?"

Because they answer different questions, they can sometimes point in different directions.

NPV is the present value of future cash inflows minus the present value of future cash outflows. It converts all future project cash flows into today's money using a chosen discount rate.

Decision rule

Accept the project if $NPV > 0$.

Reject the project if $NPV < 0$.

If projects are mutually exclusive, choose the one with the higher NPV.

Why NPV is powerful?

NPV directly measures the change in wealth. If the project has a positive NPV, it means the project is expected to add value above the required return. This is why NPV is usually treated as the best single decision criterion in capital budgeting.

Solved Example

Suppose a project requires an initial investment of \$100,000 and generates the following cash inflows:

- Year 1: \$30,000
- Year 2: \$35,000
- Year 3: \$40,000
- Year 4: \$25,000

Assume the discount rate is 10%.

Step 1: Discount each inflow

- Year 1 present value: $30,000 / 1.10 = \$27,272.73$
- Year 2 present value: $35,000 / (1.10)^2 = \$28,925.62$
- Year 3 present value: $40,000 / (1.10)^3 = \$30,052.59$
- Year 4 present value: $25,000 / (1.10)^4 = \$17,075.34$

Step 2: Sum the present values

$$27,272.73 + 28,925.62 + 30,052.59 + 17,075.34 = \$103,326.28$$

Step 3: Subtract the initial investment

$$\text{NPV} = 103,326.28 - 100,000 = \mathbf{\$3,326.28}$$

Interpretation: The NPV is positive, so the project adds \$3,326.28 in present value terms and should be accepted.

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Net Present Value and Internal Rate of Return in R



R Script (<https://learn.usc.edu.au/courses/35359/files/3226185?wrap=1>) 
(https://learn.usc.edu.au/courses/35359/files/3226185/download?download_frd=1)

How you can do this in R

The **FinCal** package makes financial calculations in R remarkably straightforward. Functions like **npv()** and **irr()** replace what would otherwise require several manual steps with a single line of code, making it easy to verify calculations and test different scenarios quickly.

```
install.packages("FinCal") # Install the package
```

```
library(FinCal) # Load the package
```

```
# Define the cash flows
```

```
# Initial investment is negative (money going out)
```

```
# Inflows are positive (money coming in)

cf <- c(-100000, 30000, 35000, 40000, 25000)

# Discount rate

r <- 0.10

# Compute NPV

npv(r = r, cf = cf)
```

Internal rate of return

IRR is the discount rate that makes NPV equal to zero. It is the project's implied rate of return.

Decision rule

Accept the project if $IRR > \text{required return}$.

Reject the project if $IRR < \text{required return}$.

IRR is attractive because it is expressed as a percentage, which makes it easy to compare with the firm's hurdle rate.

Solved IRR Example

Using the same project as above, we find the rate r such that:

$$100,000 = 30,000 / (1 + r) + 35,000 / (1 + r)^2 + 40,000 / (1 + r)^3 + 25,000 / (1 + r)^4$$

Interpretation: If the firm's required return is 10%, and the IRR exceeds that threshold, the project should be accepted under the IRR rule.

How you can do this in R

```
library(FinCal)

# Same cash flow vector as the NPV example

cf <- c(-100000, 30000, 35000, 40000, 25000)

irr(cf)
```

Why NPV and IRR can conflict

Although both methods are based on the same cash flows, they can rank projects differently. This happens mainly when the projects are mutually exclusive, meaning only one can be chosen.

- Different project scale

One project may be small and efficient, while another may be large and generate more total value. IRR is a percentage measure, so it can favour the small project. NPV is an absolute dollar measure, so it often favours the larger project if it creates more total wealth.

Example

Project A: invest \$10,000, earn \$3,000 annually.

Project B: invest \$100,000, earn \$20,000 annually.

Project A may have a higher IRR because the return is large relative to the investment. But Project B may have a higher NPV because it produces more total value.

IRR may prefer the smaller project.

NPV may prefer the larger project.

- Different timing of cash flows

A project with early cash inflows may have a higher IRR because it returns money quickly. Another project may have lower IRR but create more value overall because later cash flows are larger. IRR tends to Favor earlier payback, while NPV properly values all cash flows over time.

- Different cash flow patterns

Projects may have different combinations of initial cost, interim inflows, and later outflows. A project with a strong early inflow and weaker later inflow may have a higher IRR than one with a steadier stream of larger inflows. NPV, however, focuses on total present value, not just the percentage return.

- Reinvestment assumption

IRR assumes interim cash inflows are reinvested at the IRR itself. If the IRR is very high, that assumption may be unrealistic. NPV assumes reinvestment at the discount rate or cost of capital, which is usually more defensible.

- Mutually exclusive project ranking

The biggest conflict usually appears when choosing between two projects and only one can be selected. In such cases, IRR may rank the projects differently from NPV because it does not account for project scale and value contribution in the same way.

Which method should be used when they conflict?

When NPV and IRR conflict, NPV should generally be preferred.

Why NPV is preferred:

- It measures value added directly.

- It accounts for project scale.
- It uses a more realistic reinvestment assumption.
- It is consistent with the goal of maximizing shareholder wealth.

Practical rule

If there is a conflict:

- Check the NPV first.
- Give primary weight to NPV.
- Use IRR as a supplementary measure.

When comparing mutually exclusive projects, compare the incremental cash flows and the incremental NPV.

Incremental analysis

If you have to choose between Project A and Project B, compare the difference between them rather than evaluating each one in isolation. This helps identify which option adds more value.

Crossover rate

Sometimes the projects' NPV profiles cross at a certain discount rate. Below that rate, one project is better; above it, the other is better. This helps explain why rankings can reverse depending on the hurdle rate.

Clear decision message

- Use NPV as the main decision rule.
- Use IRR as a secondary check.
- Use incremental NPV when projects are mutually exclusive.
- Use caution if IRR and NPV disagree.

Why IRR may not be unique

IRR works cleanly only when the project has one initial outflow followed by a series of inflows. In many real projects, cash flows are more complicated.

Non-conventional cash flows

If the cash flow pattern changes sign more than once, the project may have:

- more than one IRR
- no IRR at all
- or an IRR that is difficult to interpret

Example of non-conventional cash flow

Suppose the project cash flows are:

Year 0: −100,000

Year 1: +250,000

Year 2: -160,000

This pattern has more than one sign change. Such cash-flow sequences can produce multiple IRRs.

Why this happens

IRR is found by solving a polynomial equation. When the cash flow sign changes more than once, the equation may have multiple mathematical roots. That means there may be more than one discount rate that sets NPV to zero.

Why this is a problem

If there is more than one IRR, which one should be used? The answer is unclear. That makes IRR unreliable for non-standard projects.

Another issue: no IRR

Some cash flow patterns do not produce a meaningful IRR at all. In such cases, IRR may fail to give a usable answer, while NPV still works normally.

Bottom line

If cash flows are conventional, IRR can be useful. If cash flows are unusual, multiple, or irregular, NPV is much more reliable.

Summary of NPV and IRR weaknesses

Weaknesses of NPV

- Depends on the discount rate.
- Relies on forecast accuracy.
- Does not automatically capture future flexibility.
- Can understate strategic value when managerial options exist.

Weaknesses of IRR

- Can conflict with NPV in mutually exclusive projects.
- Can favour small projects with high percentages rather than large projects with more value.
- Assumes reinvestment at the IRR.
- May produce multiple IRRs or no IRR for non-conventional cash flows.

These weaknesses do not make NPV or IRR useless. They show that both methods are best viewed as necessary but not sufficient tools.

Why real options analysis is needed

Real options analysis is needed because many projects are not rigid one-shot decisions. In practice, managers often have flexibility. They can wait, expand, delay, contract, abandon, or switch once more information becomes available.

Standard NPV assumes the firm commits now and passively follows the original plan. But real business decisions are dynamic. The ability to adapt has value, and that value is often ignored by traditional appraisal methods.

Core idea

A real option is the right, but not the obligation, to take a future action.

Examples include:

- the option to defer investment
- the option to expand if demand is strong
- the option to abandon if conditions worsen
- the option to stage investment
- the option to switch inputs or outputs

Why this matters

A project with negative static NPV may still be worth doing if it creates valuable flexibility.

Likewise, a project with positive NPV may be even more attractive than it first appears if it gives management future strategic choices.

Transition from NPV/IRR to real options

This is why a feasibility study should not stop with NPV and IRR when uncertainty is high. Those methods tell us whether the project is worthwhile under the forecast assumptions, but they do not fully measure the value of flexibility. In many projects, the biggest source of value is not just the current cash flow stream, but the strategic options embedded in the project itself.

Real options analysis extends traditional project appraisal by valuing flexibility. It recognizes that uncertainty is not always purely a risk to be minimized. Sometimes uncertainty creates opportunity, because management can wait for clearer information before committing or can respond to new developments after the project begins.

Types of real options

- (i) Option to defer: The firm can delay the investment until conditions improve or uncertainty reduces.
- (ii) Option to expand: If the project performs well, the firm can increase scale, enter new markets, or add capacity.
- (iii) Option to abandon: If performance is poor, the firm can shut the project down and recover some value.

(iv) Option to contract: If demand is weaker than expected, the firm can reduce the project size to limit losses.

(v) Option to switch: The firm can switch between technologies, products, markets, or input sources depending on what becomes more profitable.

(vi) Staged investment option: Rather than investing everything at once, the firm commits in phases. Each stage provides information that helps shape the next decision.

Real options example: expansion and abandonment

Suppose a company is considering launching a new electric scooter service in a city.

Base facts

- Initial investment: \$500,000
- Static NPV under current assumptions:
- Market demand is uncertain
- If demand is strong, the company can expand the fleet later
- If demand is weak, the company can abandon the project after a trial period

Traditional NPV view

If we only look at the immediate forecast, the project seems unattractive because NPV is negative. Under a rigid appraisal approach, the project would be rejected.

Real options view

Now consider the embedded flexibility.

If demand is strong, management can expand.

If demand is weak, management can limit losses by stopping further investment.

The ability to wait and learn before fully committing has value.

Suppose:

- the option to expand is worth \$120,000,
- the option to abandon protects \$30,000 of downside value.

Then the project's adjusted value becomes: $-50,000 + 120,000 + 30,000 = 100,000$

Interpretation:

Although the static NPV is negative, the real option value makes the project attractive. This means the flexibility to respond to future information changes the investment decision.

Why real options change the feasibility assessment

Real options matter because they capture value that static NPV misses.

Without real options: A project is judged only by the forecast cash flows in the base case.

With real options: A project is judged by

- the forecast cash flows
- the uncertainty around those cash flows
- the value of managerial flexibility

This is particularly important when:

- the market is volatile
- demand is uncertain
- technology is changing quickly
- the project can be staged
- the firm can exit easily if conditions deteriorate

Traditional NPV tells you whether a project is worth doing if nothing changes. Real options tell you what the project is worth because things may change and management can respond.

How to explain real options in a report

A good feasibility report should say more than “the project has positive or negative NPV.” It should also ask whether the project gives the firm useful flexibility.

A strong statement might be:

“The project has a negative static NPV under conservative assumptions. However, it provides the option to expand if demand is stronger than expected and the option to abandon if market conditions deteriorate. Because these flexibilities reduce downside risk and create upside potential, the project may be more valuable than the standard NPV estimate suggests.”

Practical decision framework

When evaluating a project, use the following sequence:

- Estimate cash flows.
 - Calculate NPV.
 - Calculate IRR.
 - Check whether the project is conventional or non-conventional.
 - If NPV and IRR conflict, rely mainly on NPV.
 - If uncertainty is high and flexibility is important, assess real options.
 - Treat real options as additional strategic value, not as a replacement for basic financial logic.
-

Final takeaway

NPV is the best single measure of value creation, IRR is a useful supporting percentage measure, and real options analysis becomes necessary when managerial flexibility and uncertainty are

important. Together, these tools give a more complete picture of project feasibility than any one method alone.